

Survival ML (random survival forest, DeepSurv)

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Fit a random survival forest and interpret its variable importance and risk stratification.
- Describe the DeepSurv architecture conceptually and when it is preferred over the forest.
- Compare against a Cox model on the same features.

Prerequisites

Cox regression and censoring.

Background

Survival analysis deals with time-to-event data in the presence of censoring. The Cox proportional-hazards model is the workhorse linear method; random survival forests (RSF) extend the idea of random forests to right-censored outcomes, using a log-rank split rule and Harrell's concordance as the default fit criterion. They handle nonlinearities and interactions automatically and are a good first nonparametric benchmark.

DeepSurv is a neural-network generalisation of the Cox partial likelihood: the linear predictor is replaced by a feed-forward network. In small biomedical datasets it rarely beats a well-tuned RSF or a Cox model with splines, but the approach scales to large cohorts and to inputs where a representation must be learned.

Evaluation of survival models uses time-integrated concordance (Harrell's C), the integrated Brier score, and — increasingly — decision curves at a clinically relevant horizon. Reporting only C on a single test set is weak evidence.

Setup

```
library(tidyverse)
library(survival)
library(randomForestSRC)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Fit an RSF on `survival::pbc` and compare to a Cox model using Harrell's concordance.

2. Approach

```
as_tibble() |>
drop_na(bili, albumin, age, edema, protime, stage) |>
mutate(status = ifelse(status == 2, 1, 0))

ggplot(d, aes(time / 365.25, fill = factor(status))) +
```

```
geom_histogram(bins = 30, alpha = 0.7, position = "identity") +
labs(x = "follow-up (years)", y = "count", fill = "event")
```

3. Execution

```
      protime + stage, data = d)
fit_rsf <- rfsrc(Surv(time, status) ~ bili + albumin + age + edema +
      protime + stage,
      data = as.data.frame(d), ntree = 500,
      importance = TRUE)
fit_rsf
```

```
  imp = as.numeric(fit_rsf$importance)) |>
  arrange(desc(imp)) |>
  ggplot(aes(reorder(var, imp), imp)) +
  geom_col() + coord_flip() + labs(x = NULL, y = "VIMP")
```

4. Check

Compare concordance on held-out data.

```
tr <- d[idx, ]; te <- d[-idx, ]
cox2 <- coxph(Surv(time, status) ~ bili + albumin + age + edema +
      protime + stage, data = tr)
rsf2 <- rfsrc(Surv(time, status) ~ bili + albumin + age + edema +
      protime + stage,
      data = as.data.frame(tr), ntree = 500)
c_cox <- survival::concordance(cox2, newdata = te)$concordance
p_rsf <- predict(rsf2, newdata = as.data.frame(te))$predicted
c_rsf <- survival::concordance(Surv(te$time, te$status) ~ p_rsf,
      reverse = TRUE)$concordance
c(cox = c_cox, rsf = c_rsf)
```

DeepSurv conceptual sketch.

```
library(torch)
# A DeepSurv-style loss computes the partial log-likelihood on the
# sorted risk set. The network output  $h(x)$  replaces the Cox linear
# predictor; the loss is  $-(\sum_{i:\text{event}} h(x_i) - \log \sum_{j \in R_i} \exp(h(x_j)))$ .
deepsurv_loss <- function(risk, time, event) {
  ord <- order(time, decreasing = TRUE)
  risk <- risk[ord]; event <- event[ord]
  log_cumsum <- torch_logcumsumexp(risk, dim = 1)
  -torch_sum(event * (risk - log_cumsum)) / torch_sum(event)
}
```

5. Report

On `survival::pbc`, a random survival forest achieved test concordance `round(c_rsf, 3)` compared with `round(c_cox, 3)` for a multivariable Cox model. Bilirubin, albumin, and age dominated VIMP. A DeepSurv network with the partial-likelihood loss would be the natural next step on a larger cohort.

On this sample size, the RSF and Cox model are close; on larger datasets with nonlinearities, the forest often pulls ahead.

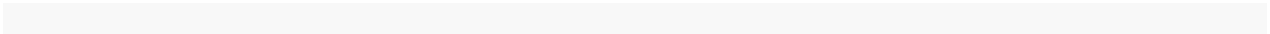
Common pitfalls

- Treating the Cox proportional-hazards assumption as always met; it often is not, and the RSF is more robust to violation.
- Comparing models without the same censoring distribution in the test set.
- Reporting only C without a calibration or decision curve at the horizon of interest.

Further reading

- Ishwaran H et al. (2008), *Random survival forests*.
- Katzman JL et al. (2018), *DeepSurv: Personalized treatment recommender system using a Cox proportional hazards deep neural network*.

Session info



See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)